

Network Technology Laboratory 2023/2024

Final project

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SUBJECT: SMALL DESIGNS FOR OFFICE NETWORK

1. THE GOAL OF THE TASK AND A SHORT DESCRIPTION OF WHAT HAS BEEN COMPLETED IN THE PROJECT:

Hello I am Firat and In this project I designed two network plan for different offices. They are two brother and they come to me to design their office network and In this project i explained what I implement for their office network.

1. I created two simple network using a Router and Access Layer Switches.
2. I connected networking devices with Correct cabling.
3. I connected Networks using dynamic routing for Brother 2 Office.
4. I did subnetting and IP Addressing.
5. I assigned IP Addresses to Router's interfaces.
6. I used static IP Address allocation to Host Devices.
7. At the end I tested and verified Network Communication

2.ADDRESSING PLAN

Network Name	Network IP	Mask	Broadcast	Gateway IP	IP of connected computers	IP of servers (what kind of servers or other devices are they
BROTHER1 Office Room 1	192.168.40.0	255.255.255.128	192.168.40.127	192.168.40.1	192.168.40.2-4	
BROTHER1 Office Room 2	192.168.40.128	255.255.255.128	192.168.40.254	192.168.40.129	192.168.40.130-132	

Network Name	Network IP	Mask	Broadcast	Gateway IP	IP of connected computers	IP of servers (what kind of servers or other devices are they
BROTHER2 Office Room 1	192.168.10.0	255.255.255.0	192.168.10.255	192.168.10.10	192.168.10.1-2	
BROTHER2 Office Room 2	192.168.11.0	255.255.255.0	192.168.11.255	192.168.11.10	192.168.11.1-2	
Office Room 3	192.168.12.0	255.255.255.0	192.168.12.255	192.168.12.10	192.168.12.1-2	

3. LIST OF DEVICES USED (type and main parameters)

3.1

INTERMEDIARY DEVICES	INTERFACE	IP ADDRESS	DEFAULTGATEWAY
BROTHER 1 Switch0 2960-24TT Switch1 2960-24TT Router0 Router-PT	Fa0/1-4	192.168.40.1	192.168.40.1
	Gig0/1 Gig0/0	192.168.40.128	192.168.40.129
BROTHER 2 Switch0 2960-24TT 2 3 Router 1 Router-PT 2 3	Fa0/1-3	192.168.10.0 192.168.11.0 192.168.12.0	192.168.10.10 192.168.11.10 192.168.12.10
	Se2/0 Se3/0	12.0.0.1 11.0.0.1 10.0.0.1	

BROTHER 1

END DEVICES	IPv4 Address	Subnet Mask	Default Gateway	INTERFACE
PC-PT (PC1)	192.168.40.130	255.255.255.128	114.0.0.129	Fa0/2
PC-PT (PC2)	192.168.40.131	255.255.255.128	114.0.0.129	Fa0/3
PC-PT (PC0)	192.168.40.2	255.255.255.128	192.168.40.1	Fa0/2
LAPTOP-PT (LAPTOP0(2))	192.168.40.3	255.255.255.128	192.168.40.1	Fa0/3
PRINTER- PT(PRINTER1)	192.168.40.132	255.255.255.128	192.168.40.129	Fa0/4
PRINTER- PT(PRINTER0)	192.168.40.4	255.255.255.128	192.168.40.1	Fa0/4

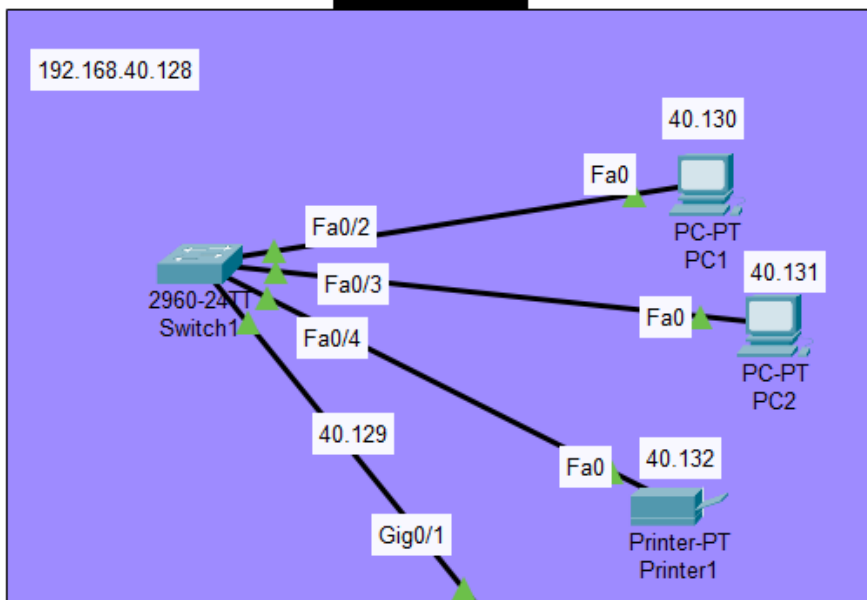
BROTHER 2

END DEVICES	IPv4 Address	Subnet Mask	Default Gateway	INTERFACE
PC-PT PC0(1)	192.168.10.1	255.255.255.0	192.168.10.10	Fa0/2
PC-PT PC1(1)	192.168.10.2	255.255.255.0	192.168.10.10	Fa0/1
PC-PT PC2(1)	192.168.11.1	255.255.255.0	192.168.11.10	Fa0/2
PC-PT PC3	192.168.11.2	255.255.255.0	192.168.11.10	Fa0/3
PC-PT PC4	192.168.12.1	255.255.255.0	192.168.12.10	Fa0/2
PC-PT PC5	192.168.12.2	255.255.255.0	192.168.12.10	Fa0/1

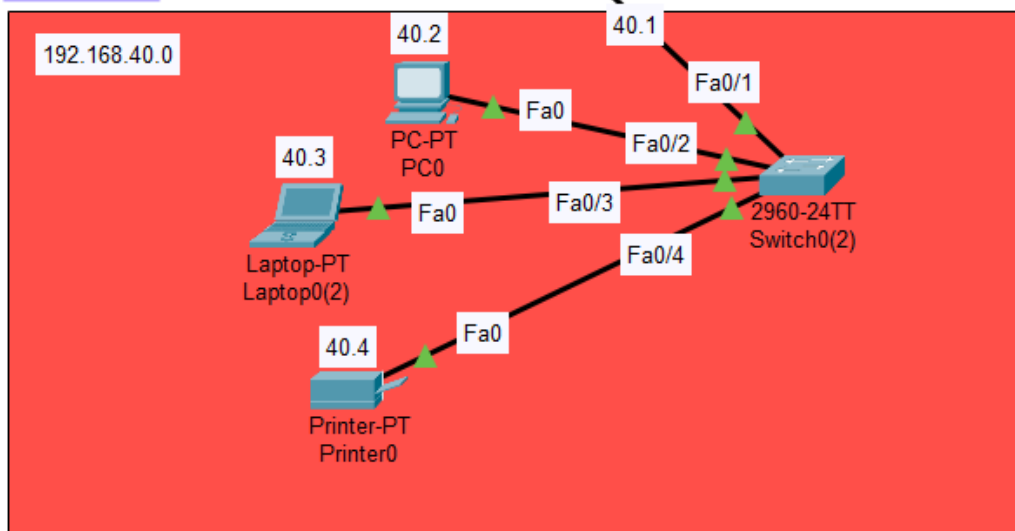
4. LOGICAL DIAGRAM OF THE NETWORK.A DESCRIPTION OF THE DEVICE CONFIGURATION

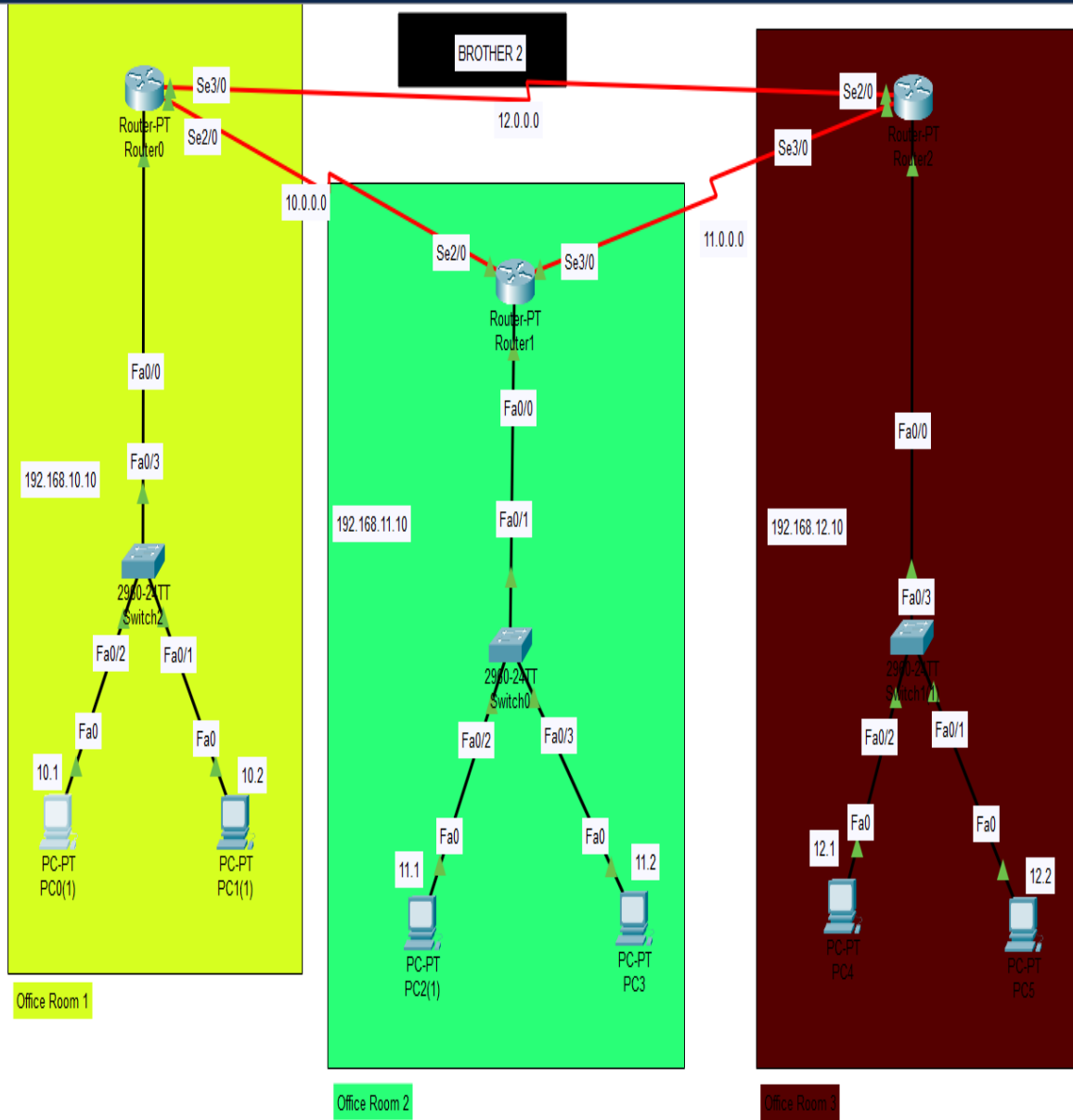
Office Room 2

BROTHER 1



Office Room 1





6. TEST OF COMPLETED NETWORK

BELOW pictures show the DHCP, DNS, HTTP, FTP services and their tests.

The image displays three screenshots of a network simulation environment, likely Packet Tracer, showing the configuration and testing of various services across a multi-branch network.

Top Screenshot: DHCP Configuration

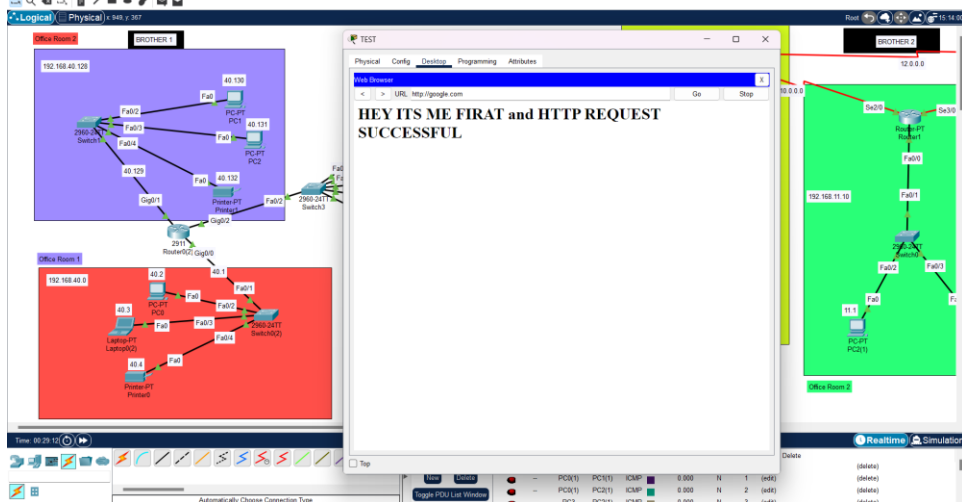
- The main window shows the network topology with three branches: Office Room 1 (purple), Office Room 2 (red), and Office Room 3 (green).
- The **TEST** window is open, showing the configuration for the DHCP service on the **Router92** (Gig0/0).
- The **IP Configuration** tab is selected, showing the DHCP service is enabled and successful.
- The **Static** configuration is shown for the DHCP service, with the following details:
 - Interface: FastEthernet0/0
 - IP Address: 192.168.55.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.55.1
 - DNS Server: 0.0.0.0

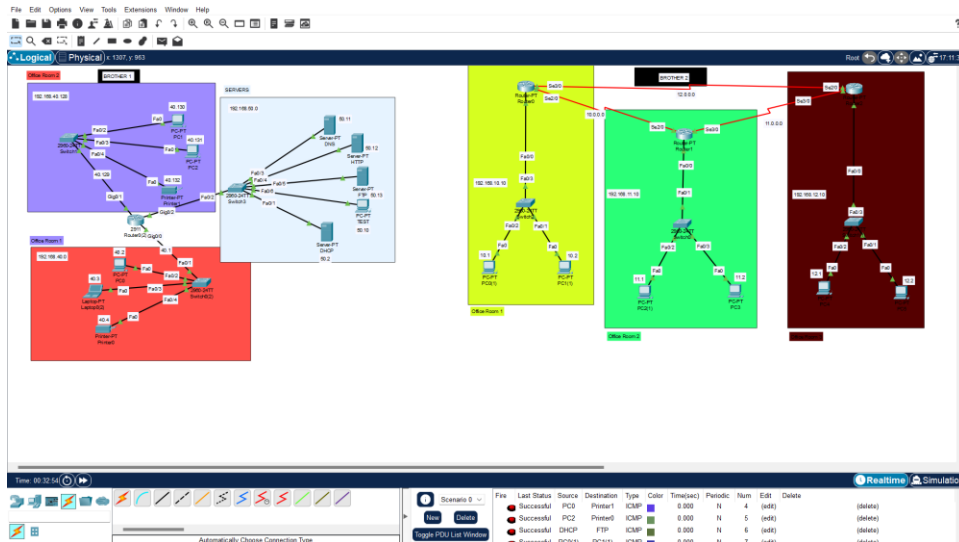
Middle Screenshot: DNS Configuration

- The main window shows the same network topology.
- The **TEST** window is open, showing the configuration for the DNS service on the **Router92** (Gig0/0).
- The **Static** configuration is shown for the DNS service, with the following details:
 - Interface: FastEthernet0/0
 - IP Address: 192.168.55.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.55.1
 - DNS Server: 192.168.55.5

Bottom Screenshot: HTTP and FTP Services

- The main window shows the same network topology.
- The **TEST** window is open, showing the configuration for the HTTP and FTP services on the **Router92** (Gig0/0).
- The **Static** configuration is shown for the HTTP and FTP services, with the following details:
 - Interface: FastEthernet0/0
 - IP Address: 192.168.55.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.55.1
 - DNS Server: 192.168.55.5





BROTHER 1 TESTS

PDU List Window										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP	Blue	0.000	N	0	(edit)	(delete)
	Successful	PC2	Printer1	ICMP	Green	0.000	N	1	(edit)	(delete)
	Successful	PC0	Laptop0(2)	ICMP	Cyan	0.000	N	2	(edit)	(delete)
	Successful	Laptop...	Printer0	ICMP	Grey	0.000	N	3	(edit)	(delete)

BROTHER 2 TESTS

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC5	PC1(1)	ICMP	Dark Blue	0.000	N	8	(edit)	(delete)
	Successful	PC1(1)	PC5	ICMP	Blue	0.000	N	9	(edit)	(delete)
	Successful	PC0(1)	PC2(1)	ICMP	Green	0.000	N	10	(edit)	(delete)
	Successful	PC0(1)	PC3	ICMP	Dark Green	0.000	N	11	(edit)	(delete)
	Successful	PC5	PC2(1)	ICMP	Blue	0.000	N	12	(edit)	(delete)
	Successful	PC0(1)	PC3	ICMP	Dark Blue	0.000	N	13	(edit)	(delete)

6. CONCLUDING THOUGHTS

In conclusion, the successful completion of this project marks a significant achievement in designing and implementing efficient office networks for two brothers, creating a solid foundation for their respective workplaces. By crafting two distinct network plans, I addressed the unique requirements of each office, ensuring optimal connectivity and functionality.

The integration of routers and access layer switches, coupled with meticulous cabling, has resulted in a robust infrastructure capable of handling the networking demands of both offices. The decision to employ dynamic routing for Brother 2's office enhances flexibility and adaptability, facilitating seamless communication within the network.

Furthermore, the implementation of subnetting and IP addressing, along with the assignment of IP addresses to router interfaces and static IP allocation for host devices, contributes to the organization and efficiency of the network. This meticulous approach not only

ensures a streamlined addressing scheme but also enhances security and management.

The concluding phase involved rigorous testing and verification of network communication. This crucial step confirms the reliability and effectiveness of the designed networks, validating that they meet the specified requirements and operate as intended.

Overall, the successful execution of this project underscores the importance of careful planning, technical expertise, and attention to detail in creating functional and reliable office networks. As Firat, you have played a pivotal role in providing tailored solutions to meet the specific needs of your clients, laying the groundwork for efficient and seamless business operations in their respective offices.